

Baseline Evaluation - Classification Reports:

	File_Name	Label	LOC	Cyclomatic_Complexity	Num_Variables
0	NestableError.java	1	179	18	3
1	ExceptionUtils.java	1	396	43	15
2	ExceptionTestSuite.java	1	89	3	2
3	StringUtils.java	1	1422	224	37
4	StringUtilsTest.java	1	287	21	13

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Logistic Regression

	precision	recall	f1-score	support
Clean (0)	0.54	0.08	0.15	224
Defective (1)	0.57	0.94	0.71	289
accuracy			0.57	513
macro avg	0.56	0.51	0.43	513
weighted avg	0.56	0.57	0.47	513

Decision Tree

	precision	recall	f1-score	support
Clean (0)	0.41	0.50	0.45	224
Defective (1)	0.53	0.44	0.48	289
accuracy			0.47	513
macro avg	0.47	0.47	0.47	513
weighted avg	0.48	0.47	0.47	513

Random Forest

	precision	recall	f1-score	support
Clean (0)	0.44	0.42	0.43	224
Defective (1)	0.56	0.58	0.57	289
accuracy			0.51	513
macro avg	0.50	0.50	0.50	513
weighted avg	0.51	0.51	0.51	513

SVM

	precision	recall	f1-score	support
Clean (0)	0.49	0.33	0.39	224
Defective (1)	0.59	0.74	0.65	289
accuracy			0.56	513
macro avg	0.54	0.53	0.52	513
weighted avg	0.55	0.56	0.54	513

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Naïve Bayes

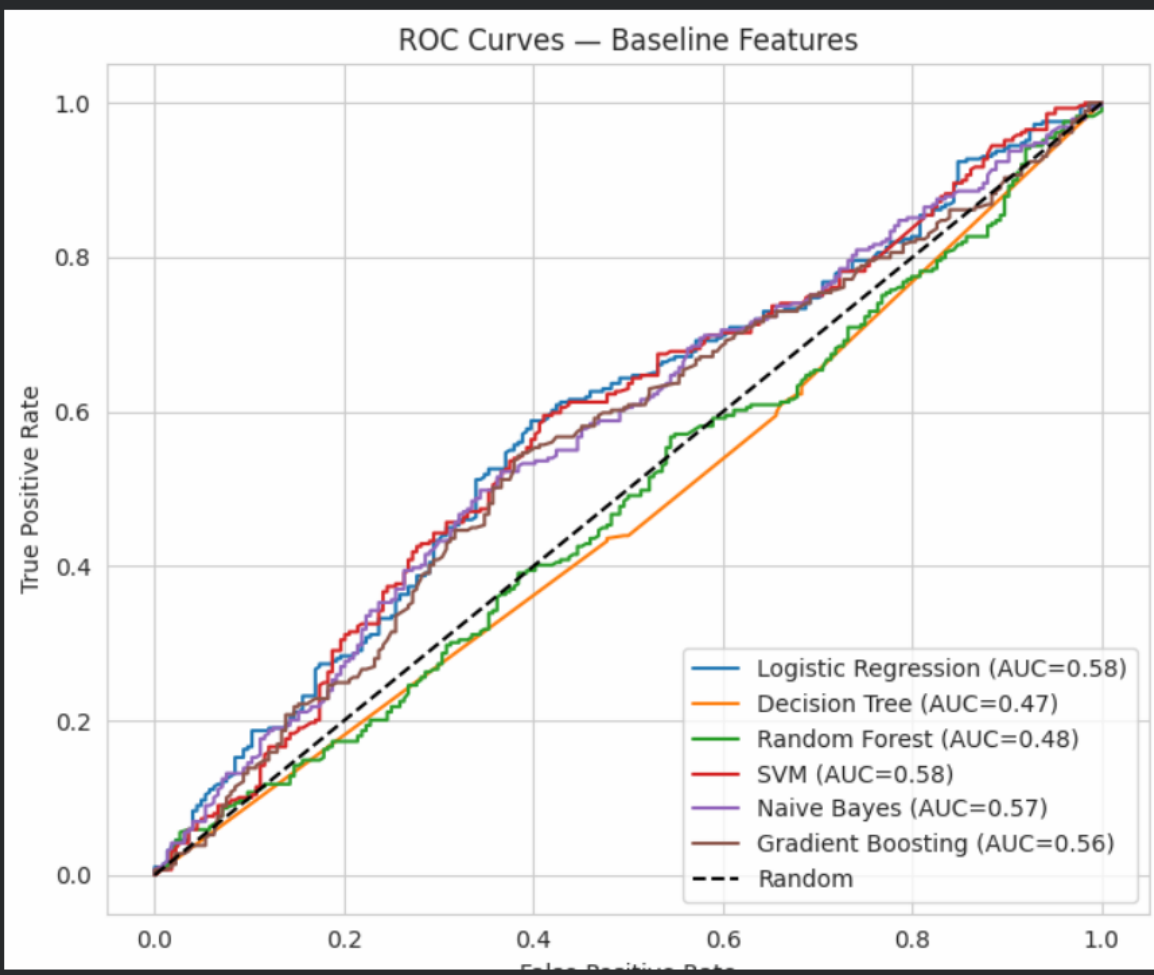
	precision	recall	f1-score	support
Clean (0)	0.45	0.84	0.59	224
Defective (1)	0.64	0.21	0.32	289
accuracy			0.49	513
macro avg	0.54	0.53	0.45	513
weighted avg	0.56	0.49	0.44	513

Gradient Boosting

	precision	recall	f1-score	support
Clean (0)	0.49	0.33	0.40	224
Defective (1)	0.59	0.73	0.65	289
accuracy			0.56	513
macro avg	0.54	0.53	0.52	513
weighted avg	0.54	0.56	0.54	513

Evaluation - F1 ,Precision, Recall , AUC-ROC Curve:

	Model	F1-Score	Precision	Recall	AUC-ROC
0	Logistic Regression	0.711864	0.571130	0.944637	0.583331
1	SVM	0.654434	0.586301	0.740484	0.579446
2	Naive Bayes	0.316883	0.635417	0.211073	0.573329
3	Gradient Boosting	0.650231	0.586111	0.730104	0.559032
4	Random Forest	0.572402	0.563758	0.581315	0.482900
5	Decision Tree	0.481061	0.531381	0.439446	0.469221



After Adding the EXPANDED_FEATURES: 'Halstead_Volume',
 'Halstead_Difficulty', 'Halstead_Effort', 'Comment_Density', 'WMC',
 'CBO', 'LCOM', 'Code_Churn', 'File_Age_Days', 'Fix_History_Count':

	Model	F1-Score	Precision	Recall	AUC-ROC
0	Gradient Boosting	0.752381	0.695015	0.820069	0.742276
1	Random Forest	0.670139	0.672474	0.667820	0.679591
2	Decision Tree	0.698413	0.712230	0.685121	0.663989
3	SVM	0.683812	0.607527	0.782007	0.637203
4	Naive Bayes	0.344473	0.670000	0.231834	0.609074
5	Logistic Regression	0.670827	0.610795	0.743945	0.606170

Baseline vs Expanded Side by side Comparison:



Random Forest Feature Importance:

